

Algorithms and Design Project

Student Handout

Project Overview

This project aims to help students design, analyze, and implement algorithms to solve real-world or theoretical problems.

Objectives

- Apply algorithm design techniques (e.g., greedy, dynamic programming)
- Analyze time and space complexity
- Implement algorithm solutions
- Compare different approaches
- Present findings clearly

Project Requirements

1. Problem Definition

- Define the problem clearly
- Identify inputs and outputs

2. Algorithm Design

- Provide pseudocode or flowchart
- Identify design technique used

3. Implementation

- Use any programming language
- Write clean and well-documented code

4. Analysis

- Time complexity
- Comparison with alternatives

5. Testing

- Include test cases
- Show results and edge cases

Deliverables

- Written report
- Source code
- Presentation (RECORDED)
- Sample inputs/outputs

Possible Project Topics

Sorting and Searching:

- Compare sorting algorithms
- Hybrid sorting design

Graph Algorithms:

- Shortest path (Dijkstra, A*)
- Minimum spanning tree

Dynamic Programming:

- Knapsack problem
- Longest common subsequence

Greedy Algorithms:

- Activity selection
- Huffman coding

Backtracking:

- N-Queens
- Sudoku solver

Real-world Applications:

- Route optimization
- Scheduling systems
- Traffic simulation

Evaluation Criteria

- Problem complexity – 20%
- Algorithm design – 25%
- Implementation – 20%
- Analysis – 15%
- Documentation – 10%
- Creativity – 10%

Submission Guidelines

- Submit before the deadline
- Ensure originality
- Properly cite references (acm)
- Follow formatting instructions (TO FOLLOW)